MIL-S-19500/100A(SigC)
2 July 1962
SUPERSEDING
MIL-S-19500/100(SigC)
30 November 1959

MILITARY SPECIFICATION

TRANSISTOR, PNP, GERMANIUM
TYPE 2N537

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for a germanium, PNP, transistor for use in large signal RF oscillator and amplifier applications, and having the following particular characteristics at T_C = + 25° ± 3°C. (See 3.2 and Table I herein.):

	P.G	P_o (Osc)	h_{fe}	h_{fb}	C_{ob}	r_b	θ_{J-A}
	db	mW	---	---	uuf	ohms	°C/mW
Minimum	10	75	10	0.90	---	---	---
Maximum	---	---	---	0.995	2.8	80	0.30

1.2 Maximum ratings. -

P_C 1/	BV_{CBO}	BV_{EBO}	I_C	T_J	T_{stg}
mW	Vdc	Vdc	uAdc	°C	°C
225	-30	-1	-100	+100	-65 to + 100

1/ This collector power dissipation is for 1000 hours expected life at T_C = + 25°C. For power ratings at T_C = + 25°C to + 100°C, derate at 3.0 mW/°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500	Semiconductor Devices, General Specification For

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be specified when requesting copies.)

3. REQUIREMENTS

3.1 General. - Requirements for the transistor shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols. - The abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

P_C ..... collector power dissipation
P.G ..... large-signal power gain
P_o (Osc)..... oscillator power output

3.3 Design and construction. - The transistor shall be of the design, construction, and physical dimensions specified in Figure 1 herein.

3.3.1 Lead arrangement. - The lead arrangement on the transistor shall be as indicated in Figure 1 herein.

3.3.2 Operating position. - The transistor shall be capable of proper operation in any position.

3.4 Performance characteristics. - The transistor performance characteristics shall be as specified in Tables I, II, and III herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

3.5 Marking.- The transistors shall be marked in accordance with Specification MIL-S-19500 and as follows. In instances where the diminutive size or lack of suitable surface area on the device would prevent a marking accomplishment readable by the unaided eye, 20/20 vision, at eight inches distance from the device, such marking may be omitted directly on the device. All required marking shall be placed on the unit package.

3.5.1 Type-designation marking.- Type-designation marking of transistors procured on Department of Army contracts, and which have passed Government inspection and comply with all requirements of this specification, shall consist of: "USA-manufacturer's qualification code letters - transistor designation." The letters "JAN" or any abbreviation thereof shall not be used. If any specification waiver has been granted, the combination "USA - manufacturer's qualification code letters" shall not be used to complete the type-designation marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and Acceptance Inspection.- Qualification and Acceptance Inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Group A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II, and III respectively, herein. Group C inspection shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. Acceptance Inspection shall include inspection of Preparation For Delivery (see 5.1 herein).

4.2.1 Specified LTPD for subgroups.- The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

4.2.1.1 Sample size for subgroups.- The sample size shall be selected by the manufacturer and shall be within the Max. Acc. No. limit specified herein relative to the particular subgroup LTPD.

4.2.2 Reinspection.- For customer verification inspection, lots may not be rejected unless the sample size for the particular subgroup being reinspected is not smaller than that required for the Max. Acc. No. specified herein relative to the subgroup LTPD.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

4.2.3 Disposition of sample units subjected to destructive tests.- Sample units subjected to Group B, Subgroups 2, 3, and 4 tests, and to Group C tests shall not be delivered on the contract or order.

4.3 Particular examination and test procedures.-

4.3.1 Base Spreading Resistance Test.- The test objective and results will apply to the following considerations. The base spreading resistance represents the resistance of the bulk material from which the base is made, and appears in series with the base lead. Its magnitude depends on the thickness of the base and the resistivity of the base material.

4.3.2 Oscillator Power Output test.- The output power shall be measured in accordance with the test circuit and under the conditions specified in Figure 2A, herein.

4.3.3 Large-Signal Power Gain test.- The power gain shall be measured in accordance with the test circuit and under the conditions specified in Figure 2B, herein.

4.3.4 Lead Fatigue test.- Lead shall be restricted such that lead bend shall start no closer than 3/32 ± 1/32 inch from the body of the transistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

Table I. Group A Inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup I		10	4				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
3206	Small-signal short-circuit forward-current transfer ratio (common base)	V_{CB} = -10 Vdc I_E = 10 mAdc f = 1 kc			h_{fb}	0.90	0.995	---
3266	Base spreading resistance	V_{CB} = -10 Vdc I_E = 10 mAdc f = 250 Mc 1/			r_b	---	80	ohms
3206	Small-signal short-circuit forward-current transfer ratio (common emitter)	V_{CE} = -10 Vdc I_E = 10 mAdc f = 100 Mc			h_{fe}	10	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

Table I. Group A Inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3		5	3				
3036	Collector-base cutoff current	Bias Cond. D V_{CB} = -20 Vdc			I_{CBO}	---	-5	uAdc
3001	Breakdown voltage, collector-to-base	Bias Cond. D I_C = -100 uAdc			BV_{CBO}	-30	---	Vdc
3026	Breakdown voltage, emitter-to-base	Bias Cond. D I_E = -1 mAdc			BV_{EBO}	-1	---	Vdc
3071	Collector-emitter saturation voltage	I_C = -70 mAdc I_B = 20 mAdc			V_{CE} (sat)	---	-4.5	Vdc
3236	Output capacitance	V_{CB} = -10 Vdc I_E = 0			C_{ob}	---	2.8	uuf
	Subgroup 4		5	3				
---	Oscillator power output	2/			P_o (Osc)	75	---	mW
3255	Large-signal power gain	3/			P.G.	10	---	db
3126	Thermal resistance	---			θ_{J-A}	---	0.30	°C/mW

1/ See 4.3.1 herein. 2/ See 4.3.2 herein 3/ See 4.3.3 herein

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		20	3				
2066	Physical dimensions	---			---	---	---	---
	Subgroup 3		10	4				
2031	Soldering heat	---			---	---	---	---
1051	Temperature cycling	Test Cond. B $T_{(high)}$ = + 100°C			---	---	---	---
1021	Moisture resistance	---			---	---	---	---
	End-point tests:							
3206	Small-signal short-circuit forward-current transfer ratio (common base)	V_{CB} = -10 Vdc I_E = -10 mAdc f = 1 kc			h_{fb}	0.89	---	---
3036	Collector-base cutoff current	V_{CB} = -20 Vdc			I_{CBO}	---	-6	uAdc
	Subgroup 3		20	4				
2016	Shock	Non-operating G = 500 5 blows of 1 msec ea. in orientations X1, Y1, Y2, Z1 (total = 20 blows)			---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection- (Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3 (Cont'd)							
2006	Constant acceleration (centrifuge)	G = 10,000 Orientations X1, Y1, Y2, Z1			---	---	---	---
2046	Vibration fatigue	Non-operating			---	---	---	---
2056	Vibration, variable frequency	---			---	---	---	---
	End-point tests:							
	Same as for Subgroup 2, above							
	Subgroup 4		20	3				
2036	Lead fatigue	Test Cond. E Weight =16 ±1 oz. 1/			---	---	---	---

1/ See 4.3.4 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

Table III. Group C inspection 1/

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		λ = 10	---				
1031	High-temperature life (non-operating)	T_{Stg} = 100°C			---	---	---	---
	End-point tests:							
3206	Small-signal short-circuit forward-current transfer ratio (common base)	V_{CB} = -10 Vdc I_E = -10 mAdc f = 1 kc			h_{fb}	0.89	---	---
3036	Collector-base cutoff current	V_{CB} = -20 Vdc			I_{CBO}	---	-10	uAdc
3001	Breakdown voltage, collector-to-base	Bias Cond. D I_C = -100 uAdc			BV_{CBO}	-30	---	Vdc
3071	Collector-emitter saturation voltage	I_C = -70 mAdc I_B = 20 mAdc			V_{CE} (sat)	---	-5.0	Vdc

1/ See 4.2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. - Preparation for delivery shall be in accordance with Specification MIL-S-19500.

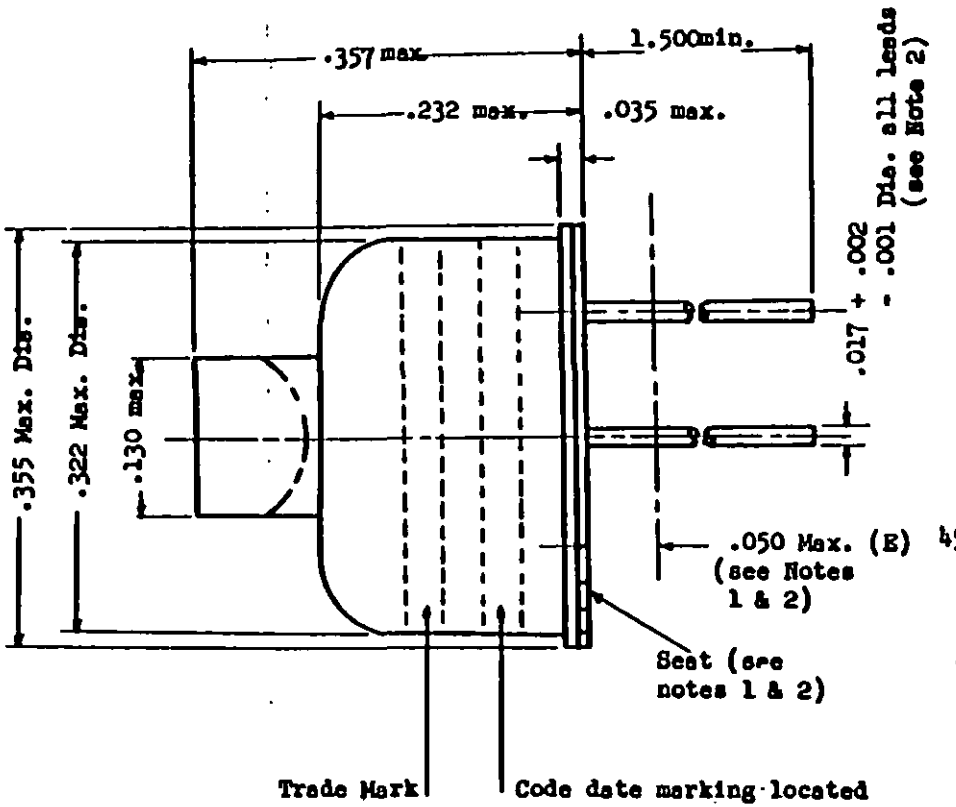
6. NOTES

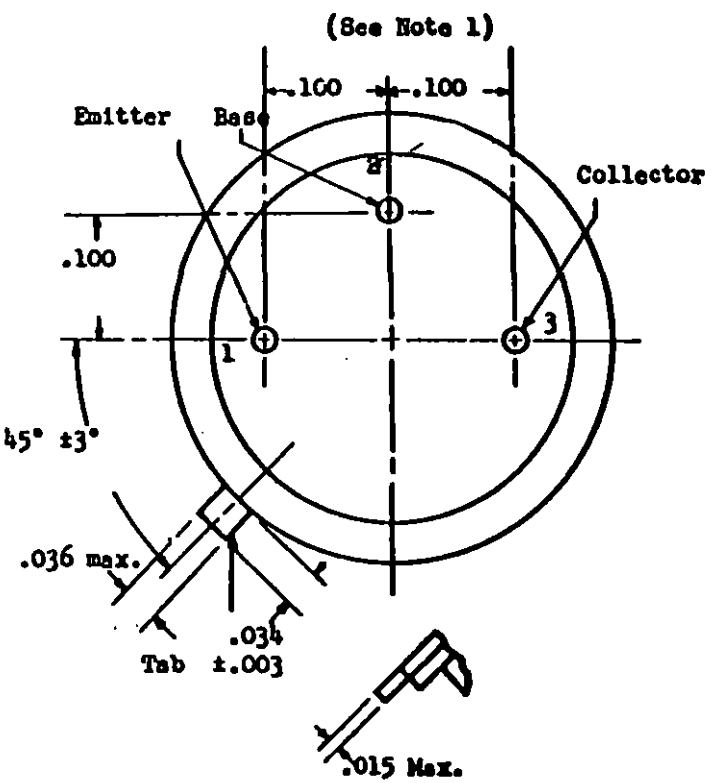
6.1 Notes. - The notes included in Specification MIL-S-19500, with the following exception, are applicable to this specification.

6.2 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL) -19500, supplement (Army), whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Chief, Standardization Division, U. S. Army Signal Materiel Support Agency, Fort Monmouth, New Jersey, Attention: SIGMS-PSM-3.

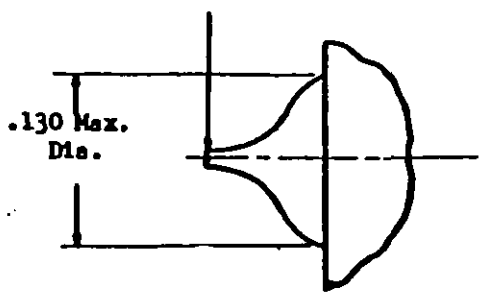
NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.





Angular orientation of edge with respect to tab optional

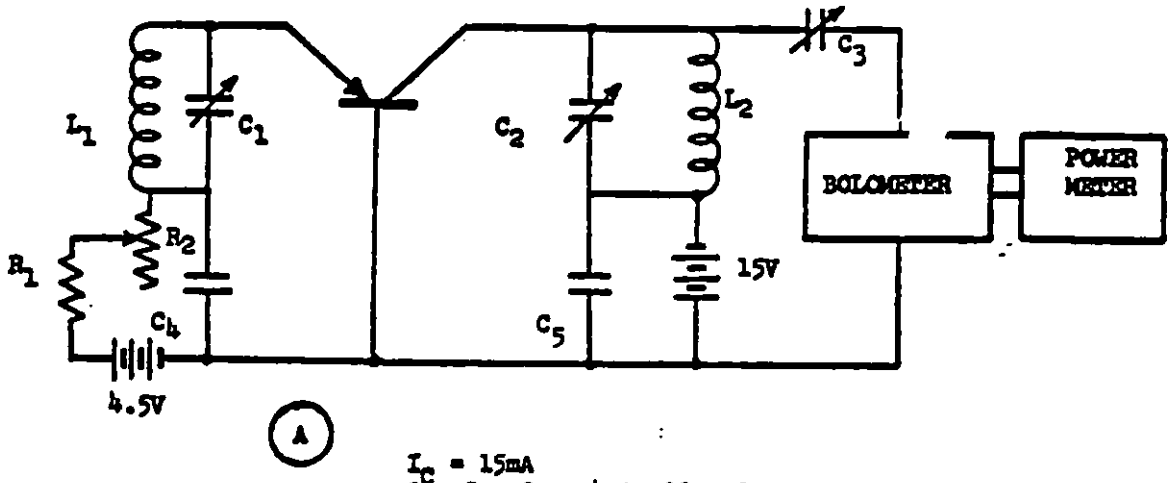


NOTES:

1. Max. dia. leads shall be within .010 of their true centers with respect to the .355 dia. measured at points .050 max. below the seat.
2. The specified lead diameter applies in the zone between .050 and .250 from the seat. Between .250 and 1.500 the diameter shall not exceed .021. Outside of these zones the diameter is not controlled.

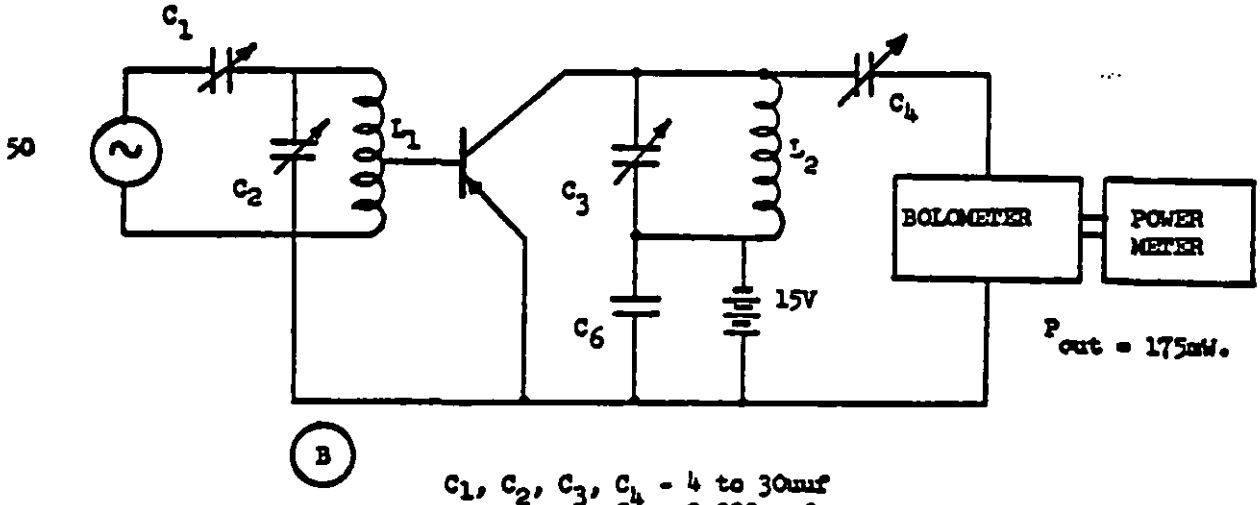
Figure 1 - Outline and Dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.



I_C = 15mA
C_1, C_2, C_3 - 4 to 30 uuf
C_4, C_5 - 2,000 uuf
L_1, L_2 - .037 uh
R_1 -220 , 1½ watt
R_2 -250 potentiometer

200 Mc Oscillator



C_1, C_2, C_3, C_4 - 4 to 30uuf
C_6 - 2,000 uuf
L_1 - .30 uh CT
L_2 - .30uh

70 Mc Amplifier

Figure 2. Oscillator Power Output and Power Gain test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications - Electronics Command
ATTN: DRSEL-ED-TO
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FOR..., CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:11Z
Check the source to verify that this is the current version before use.